Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Triple T
Project / Solution Name	GreenReady
Name (Leader)	Remy Ong Jing Yi
Email & Phone (Leader)	remyongjingyi1233@gmail.com / +60126406646
University Affiliation(s)	Asia Pacific University of Technology & Innovation
Live Prototype URL	https://sdc-hackathon-triple-t.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/RemyOng/SDC_Hackathon_Triple-T

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

GreenReady provides an AI-assisted document-processing workflow for Malaysian SMEs. An SME submits an electricity bill or renewable-energy quotation through WhatsApp, and the WhatsApp webhook is received by a Python FastAPI backend hosted on Replit. The backend retrieves the uploaded document through Whapi.cloud and stores the original file and document metadata in Supabase Storage and Supabase Postgres. The document is converted into images and sent to a Qwen2-VL vision-language model deployed through a RunPod Serverless Endpoint. The extracted fields are validated using Pydantic before deterministic Python calculations estimate annual energy generation, energy coverage, electricity savings, carbon emissions avoided, return on investment, and payback period. LangGraph manages the workflow and pauses execution for human approval. The SME receives the calculated results through WhatsApp and must explicitly reply with an approval message before the anonymised dossier is published to the marketplace database. The planned web frontend uses HTML, CSS, and JavaScript and communicates with the FastAPI backend through JSON APIs. Gmail API integration is also used to monitor and process incoming bill emails.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python	Backend application and data-processing logic	Python Software Foundation License
FastAPI	HTTP API and webhook server	MIT
LangGraph	Workflow orchestration and human-approval checkpoint	MIT
Pydantic	Validation of AI-extracted document fields	MIT
Supabase Python Client	Database and object-storage access	MIT
Google API Python Client	Gmail API integration	Apache-2.0

Name & Version	Purpose in Project	License
Google Auth / Google Auth OAuthlib	Gmail OAuth authentication	Apache-2.0
PyMuPDF	PDF and document conversion before AI processing	AGPL-3.0 or commercial license
Requests	HTTP communication with external services	Apache-2.0
python-dotenv	Loading environment variables locally	BSD-3-Clause
HTML5, CSS3, JavaScript	Planned web frontend	Standards / project-authored code

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless API	Sends document images to the deployed Qwen2-VL inference endpoint for structured extraction	Paid or credits-based; confirm actual plan
Supabase REST API	Stores SME profiles, incoming document records, extracted results, audit events, and marketplace listings	Free or paid; confirm plan
Supabase Storage API	Stores uploaded electricity bills and vendor quotations	Free or paid; confirm plan
Whapi.cloud API	Receives WhatsApp webhooks, downloads incoming media, and sends result and approval messages	Paid or trial; confirm plan
Gmail API	Reads incoming emails and attachments containing electricity bills or vendor quotations	Google API / account-dependent
Google OAuth 2.0	Authenticates the Gmail integration	No separate API charge normally

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Hosts the FastAPI backend and planned HTML/CSS/JavaScript frontend	Replit plan and deployment region to be confirmed
RunPod	Hosts the GPU inference worker and Qwen2-VL model	Serverless endpoint; region/plan to be confirmed
Supabase	Managed Postgres database, object storage, and REST access	Project region and plan to be confirmed
Whapi.cloud	WhatsApp messaging integration and webhook delivery	Provider plan to be confirmed

Service & Provider	Purpose in Project	Plan / Region
Google Cloud	Gmail API and OAuth credentials	Google Cloud project; billing status to be confirmed

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen2-VL — exact checkpoint/version to be confirmed	RunPod Serverless Endpoint	Pre-trained vision-language model used to extract structured fields from electricity bills and vendor quotations	Declare the licence for the exact Qwen2-VL checkpoint used
LangGraph workflow engine	LangChain ecosystem	Workflow orchestration, not an AI model	MIT

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
None — no training or fine-tuning dataset used	Not applicable	Not applicable	Not applicable	No
User-submitted electricity bills and vendor quotations	Submitted by prototype users through WhatsApp or Gmail	Private / self-collected operational documents	Owned or controlled by the submitting user; used with consent	Potentially Yes

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

■ **Yes** ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude 5.5 Sonnet	UI	Moderate
GPT - 5.6 Luna	Code suggestions, debugging assistance, documentation, and implementation support	Moderate
Gemeni 3.1 Pro	Brainstorming, code review, documentation, and troubleshooting	Minor or Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team reviewed AI-generated outputs using structured JSON validation, Pydantic schema checks, confidence thresholds, and deterministic business rules. Extracted bill values were cross-checked against the original documents, including usage, charges, meter readings, and billing dates. Outputs that were incomplete, inconsistent, or below the confidence threshold were flagged for human review rather than being accepted automatically. The team remains responsible for validating AI results, correcting errors, and ensuring that recommendations are calculated using deterministic rules.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

■ **Yes** ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
SDC	Replit	Used as a development and prototyping environment for building and testing the GreenReady prototype. This platform is also used as hosting as well.
SDC	Runpod	Used to host and run the vision-language model that reads and extracts information from TNB electricity bills and uploaded documents.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

RunPod was important to the prototype's automated bill-reading functionality because it enabled image and PDF documents to be converted into structured bill data. However, it could be replaced by another compatible vision-language model or cloud inference provider. Replit supported development and prototyping but was not essential to the core functionality, as the application can also run locally or on another hosting platform.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

GreenReady processes WhatsApp phone numbers, Gmail sender information, electricity bills, vendor quotations, company names, account numbers, energy-consumption values, equipment costs, and calculated financial and environmental metrics. Uploaded documents are stored in Supabase Storage, while document metadata, extracted fields, SME profiles, audit events, and marketplace listings are stored in Supabase Postgres. Document images may be sent to the RunPod-hosted Qwen2-VL inference endpoint for extraction. WhatsApp messages are sent through Whapi.cloud and email data is accessed through Gmail API with OAuth. API keys, OAuth credentials, and database secrets are stored in environment variables or platform secret storage and are not committed to the repository. The prototype requires explicit WhatsApp approval before publishing an anonymised listing. Retention periods and deletion procedures should be configured before production use; the current prototype does not yet implement a complete automated deletion policy.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The vision-language model may misread documents, hallucinate values, or fail when bills are blurry, incomplete, handwritten, or formatted unexpectedly. Model output is restricted to a Pydantic schema, but schema validation does not prove that the extracted values are factually correct. The deterministic calculator prevents the AI model from directly generating financial or carbon calculations, but the calculations remain dependent on the accuracy of extracted inputs and the assumptions used for Malaysian grid-emission and electricity-tariff factors. The prototype also depends on RunPod, Supabase, Whapi.cloud, Gmail, and Replit availability, quotas, network connectivity, and pricing. Human approval is required before marketplace publication. Further mitigation should include confidence scores, manual document review, stronger access controls, private storage URLs, rate limiting, audit-log monitoring, and automated data-retention controls.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [ ] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [ ] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [ ] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [ ] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [ ] Any use of official SDC sponsor resources has been declared in Section 5.
- [ ] The prototype described here is accessible to judges through the links provided in Section 1.
- [ ] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



24th October 2026